Session - 6

Setting Up CI/CD

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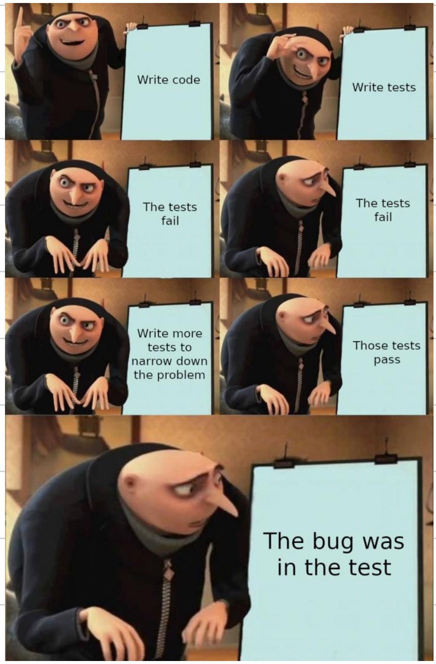
A QA engineer walks into a bar. Orders a beer. Orders 0 beers. Orders 99999999999 beers. Orders a lizard. Orders -1 beers. Orders a ueicbksjdhd.

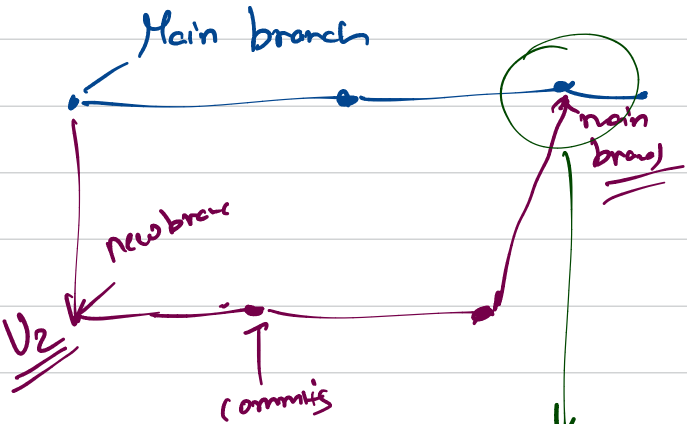
First real customer walks in and asks where the bathroom is. The bar bursts into flames, killing everyone.

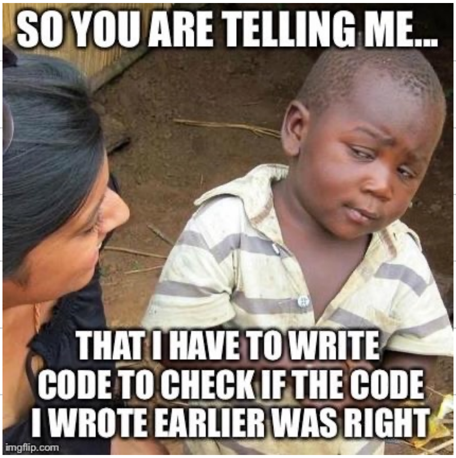


Agenda

1. Tests!! Wait what?? What's that ?!
2. What is CI/CD } As a data-scientist not allowed to work on CI/CD

TESTS





no conflict issue

merge this??

Google / Amazon → millions of microservices working together

testing goes on for months

Series of Step

1. Checkout new branch

2. Do dev & push to new branch

→ Staging Server

my code/model

GCP/Azure

Stress testing

hit your model

1900 req/min

Jmeter

randomly sample a req

hit api

1900 req/min

measure latency

> 50 ms

3 Create a pull req.

1st Project lead → review

2n Manager → review

3 Merged with main branch

Poor model training → BEWARE Fellow DS

1 Some where upstream some pipeline broken

→ Solutions

Amazon DEEQU

column

Nan Values

→ % of null-value

→ min of today's date

max

25 percentile, 75, IQR

latency → Time taken for model to respon

\+

Networking delay

Time complexity

↓

Inference level

GRPC Protocol

reduce networking latency

Inference

data → model → come out

→ Most companies prefer these:

→ Linear regression

Logistic "

Random Forest

GBM → xgboost / Lightgbm

test_file.py

def test_f1 ( ) → ✓F1 ×F2

(·) (F)

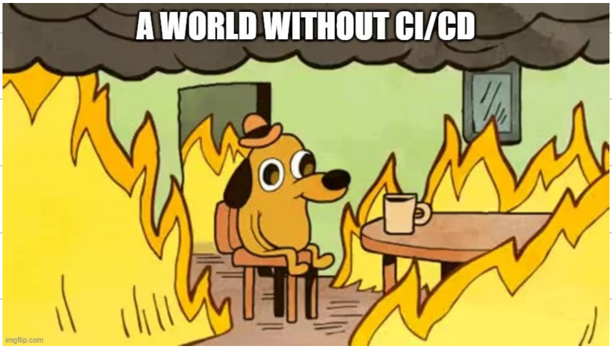
×F ✓F2

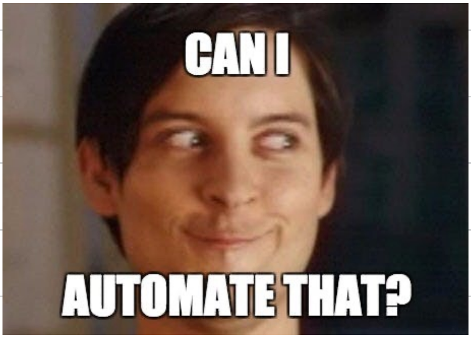
F ·

def test_ F2( ) ✓F1 ✓F2

· ·

CI/CD





Workflow → Series of steps

→ Set up Venv
→ Install depen
→ Run tests
→ Push to docker

YAML → Yet another markup Language

CI/CD → continuous deployment

→ continuous Integration

→ Github CI/CD

→ Jenkins

→ Azure CI/CD

→ Travis

→ Circle CI

→ Gitlab

→ Continuous Integration

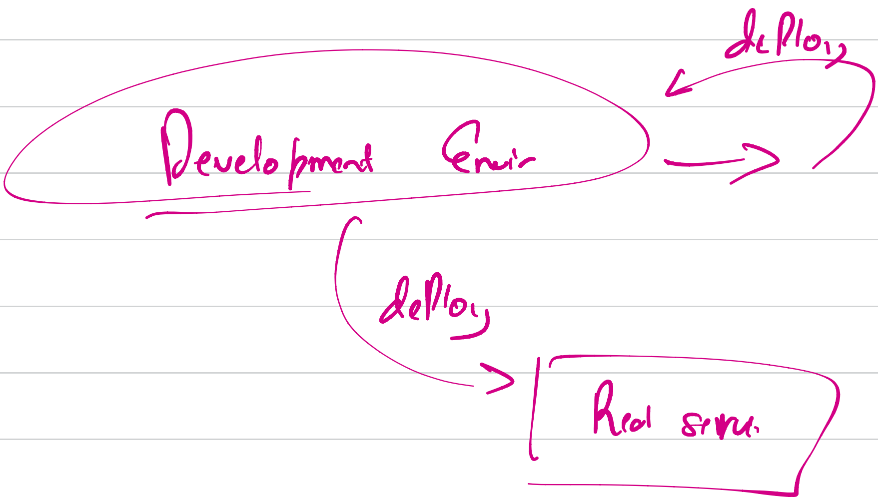
plan → code → build → test

→ CI

main → most stable release of a product

Continuous deployment

testing → deployment → monitor & operate



CI → automate test

CD → automate deployment

→ Typically jobs in workflow run parallely

Parallel making sense

Job 1 → Testing model A
Job 2 → " " B

Job1 → Testing model Flask

Job2 → pushing to docker